

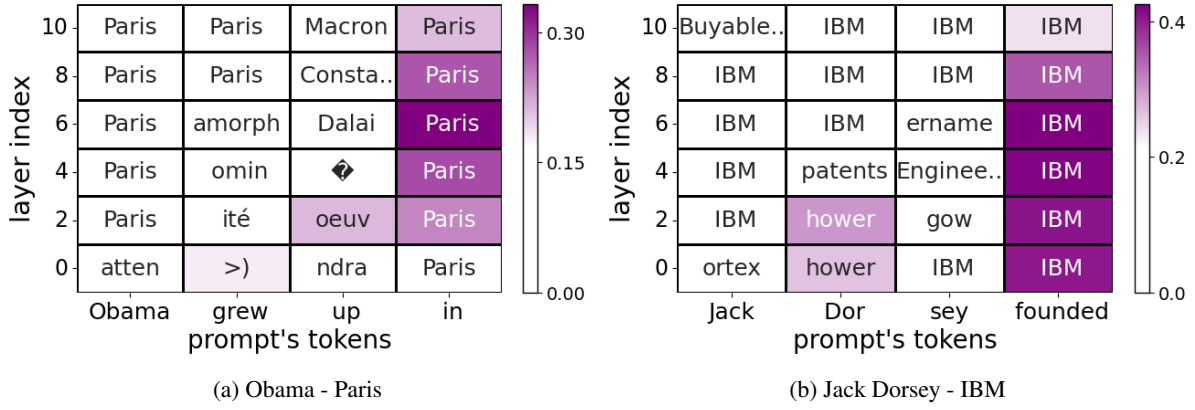


Figure 4: The gradient of GPT2-small FF_2 when (a) editing the model to answer “Paris” for the prompt “Obama grew up in”, and (b) when editing the model to answer “IBM” for “Jack Dorsey founded”. Each cell shows the Logit Lens projection of the gradient’s VJP (δ_i) for a token input and a layer. Non-English characters are replaced with a question mark, and long tokens are truncated with “..”. According to [Section 5.2](#), instead of showing the most probable token in each cell, we display the least probable one. The color indicates the norm of the VJP, with white cells indicating that almost no editing is done in practice.

in the rank of the gradients, indicating linear dependency in x_i or δ_i , see [Section 4.1](#). This is not a result of a repeated token, since the positional encoding would still lead to a different x_i .

Logit Lens of Gradients Next, we present examples of our gradients’ interpretation through LL in [Figure 4](#) and in [Appendix E](#). In each cell of these plots, the LL projections of the chosen spanning set (FF_1 ’s x_i and FF_2 ’s δ_i) are presented for a specific layer and a token from the prompt that was used for the editing.

Prior studies that projected the forward pass examine the LL projections of hidden states, highlighting the gradual change in the projected tokens between layers ([nostalgebraist, 2020](#); [Haviv et al., 2023](#)). Similarly, [Figure 4a](#) presents a gradual change in the backward pass’ VJP. Across most layers, LL reveals that the gradients represent the embedding of “Paris”. Other projections have semantics that are related to “Paris” such as “Macron”, the family name of the President of France. The norm of the VJP is indicted by color, and, in the top layers, the only meaningful updates are for the token “Paris”. Some of the edits in the lower layers are harder to explain, similarly to the situation in those layers for the vanilla LL of the forward pass. Another example, presented in [Figure 4b](#), shows similar patterns: when attempting to edit the model to suggest that Jack Dorsey, known as one of Twitter’s founders, founded IBM, we observe that most gradients correspond to the embedding of “IBM”.

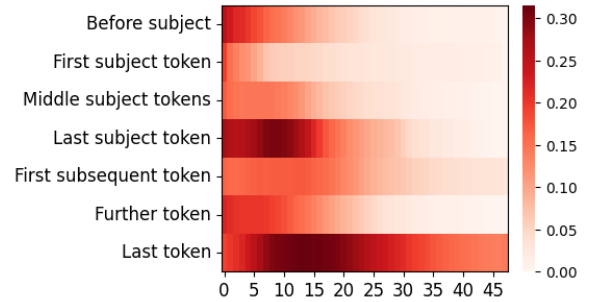


Figure 5: The norm of GPT2-xl’s FF_2 ’s VJPs (δ_i) as a function of the layers’ index and segments of the edited prompts. White color represents close-to-zero updates with almost no effect on the model’s weights.

Impact of Different Segments of the Prompt

We observe that while all the prompt’s tokens contribute to the gradient construction ([Equation 8](#)), the majority of these contributions are done by VJPs, δ_i , with a close-to-zero norm. Furthermore, upon examining the LL of every individual neuron from the gradient matrix ([Appendix C](#)), we found that all the projected tokens are correlated with only 1-2 vectors we can identify from the spanning sets presented in [Section 4.2](#).

To discern the relative importance of tokens and layers in the gradient reconstruction, we divide each prompt’s tokens into segments and plot their δ_i mean norm. This experiment is done with GPT2-xl, due to its extensive use in prior work on interpretability research.

Results are depicted for FF_2 in [Figure 5](#), see [Appendix F.1](#) for FF_1 . Evidently, predominant

updates occur in two main areas: (1) by the subject’s tokens in the initial layers, and (2) by the last prompt’s token around the second quarter of the layers. The majority of other tokens exhibit a norm close to zero throughout the layers, indicating that they have almost no effect on the updating. We hypothesize that the changes to the last subject token may involve editing the information transferred by the subject’s token through attention, as demonstrated by Geva et al. (2023).

A complementary view is provided by considering the LL rank of the target token for each VJP δ_i (labeled by the segment of token i of the input). Figure 6 illustrates that the VJP of the last token from the edited prompts, δ_n , consistently ranks the target token among the least probable ones. The VJPs of other tokens from the edited prompt, δ_i , exhibit comparable behavior, generally ranking the target token as improbable.

The result reveals that along the first and last layers, some of the δ_i show degradation in their ranking of the target token, which we attribute to their low norms as reflected by Figure 5. We demonstrate in Appendix H that normalizing δ_i before LL magnifies the presence of the target token. Specifically, the drop at the model’s last layer is due to the fact that apart from the last prompt’s token, all the others have a zero vector δ_i at that layer (Appendix A).

Please note that the degradation of this rank in the first few layers might be related to the gap in LL interpretability for the earlier layers discussed in Section 3.3. In Appendix F.2 we provide a similar analysis for FF_1 ’s gradients.

7 Application: Editing Based on the “Shift” Mechanism

Prior work (Mitchell et al., 2021; Meng et al., 2022, 2023) introduced editing methods that change only FF_2 ’s matrices. In Section 5.2 we identify the “shift” mechanism of editing FF_2 . In Section 6 we observe that the dominant components in constructing the gradients are derived from the outer product of the last token’s input and VJP, $x_n^\top \cdot \delta_n$, and that δ_n contains the embedding of the target token.

We hypothesize that we can edit LMs’ internal knowledge by updating only a single FF_2 matrix with a single forward pass and an approximation of the VJP, thereby eliminating the need for the backward pass. Based on Lemma 5.1, the embedding of the editing target is annotated by $D^\top[t]$,

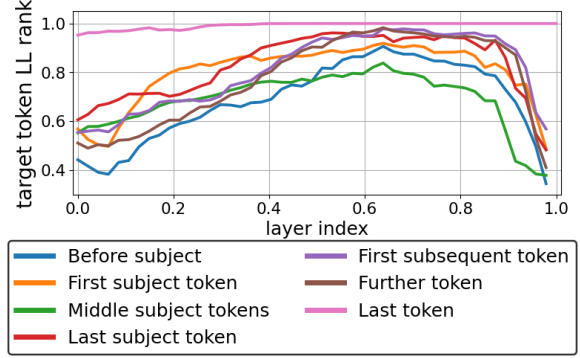


Figure 6: The Logit Lens rank of the target token for GPT2-xl FF_2 ’s VJPs, δ_i . Most gradients tend to rank the target token as one of the least probable tokens, with the last token consistently ranking it as such. We found that the degradation of the first and last layers can be attributed to the proximity of certain δ_i norms to zero. In the initial layers, Logit Lens is less effective, thereby resulting in lower readability for the earlier layers.

where D is the decoding matrix and t is the index of the target token. Our experimental method works as follows: (i) We choose an MLP layer we wish to edit, $\hat{FF}_2$ (predefined as a hyper-parameter according to Appendix I). (ii) We run a single forward pass with the prompt whose output we want to edit. (iii) During the forward pass, we collect the last token input for the layer we want to edit, x_n . (iv) We collect the embedding of the target token $D^\top[t]$ and (v) update the MLP matrix by $\hat{FF}_2 \leftarrow \hat{FF}_2 + \eta \cdot x_n^\top \cdot D^\top[t]$, where η is the learning rate. We name this method “forward pass shifting”. As motivation, please observe that $x_n(\hat{FF}_2 + \eta \cdot x_n^\top \cdot D^\top[t])D^\top[t] - x_n\hat{FF}_2D^\top[t] = \eta \cdot \|x_n\|_2^2 \cdot \|D^\top[t]\|_2^2$.

We examined our method on 1000 samples from CounterFact (Table 1; the full results and additional implementation details are presented in Appendix I), and found that for single editing our approach is on par with the state-of-the-art methods MEND (Mitchell et al., 2021), ROME (Meng et al., 2022) and MEMIT (Meng et al., 2023), in editing a given prompt, but it falls short in comparison to ROME in generalization (editing paraphrases) and specificity. However, our method has much lower runtime complexity and does not employ a multi-step (iterative) execution. Overall, our results suggest we might be able to find “shortcuts” in the implementation of fine-tuning by injecting tokens directly into LMs’ layers.

METHOD	EFFICACY $\uparrow$	PARAPHRASE $\uparrow$	NEIGHBORHOOD $\uparrow$	N-GRAM $\uparrow$
ORIGINAL MODEL	0.4	0.4	11.21	626.94
FINETUNING (MLP 0)	96.4	7.46	10.12	618.81
FINETUNING (MLP 35)	100.0	46.1	5.59	618.50
MEND	71.4	17.6	7.73	623.94
ROME	99.4	71.9	10.91	622.78
MEMIT	79.4	40.7	10.98	627.18
FORWARD PASS SHIFT	99.4	41.6	6.02	622.45

Table 1: GPT2-xl single editing results for CounterFact. Efficacy represents the editing success rate (accuracy). Paraphrase denotes the accuracy of predicting the new target for phrases derived from the edited prompt. Neighborhood examine the model accuracy on prompts we wish the edit will not change but have similar domain to the edited one. N-gram measures generation fluency using weighted bi- and tri-gram entropies.

8 Conclusions

Other LL-type interpretability contributions shed light on LMs through the forward pass. Here, we show that gradients can be projected into the vocabulary space and utilize the low-rank nature of the gradient matrices to explore the backward pass in an interpretable way. As we show, the gradients are best captured by a spanning set that contains either the input to each layer, or its VJP. These two components, which are accessible during the forward and backward passes, are used to store information in the MLP layers, using a mechanism we call “imprint and shift”. We provided experimental results to substantiate the results of our analysis, including an editing method that only requires a single forward pass, but is on par with the SOTA knowledge editing methods.

Future Work: The focus of this work is in understanding the mechanism behind the simplest form of LM editing. We hope our observations can serve further research on how knowledge is embedded into the model’s parameters. Scaling our experiments to multiple editing tasks or full fine-tuning would result in losing the low-rank characteristic that we utilize in this work. However, our findings in [Section 6](#) suggest that only a small set of VJPs and subspaces dominate the editing process. Therefore, future research could interpret complex editing scenarios by examining only the most dominant VJPs, or by decomposing the full gradients into low-rank components using techniques such as singular value decomposition (SVD).

Notably, this work did not explore the similarity between gradients of different sentences’ editing. The question of how subspaces are shared between

different edits and tasks remains an open question, which has been recently discussed in the context of superposition ([Elhage et al., 2022](#)). While most work has examined superposition phenomenon by looking at the forward pass of LMs, exploring how information is embedded into the same subspaces might shed light on how LMs work within their constrained embedding space.

In addition, in this work we utilize Logit-Lens (LL) projection, which is known to have reduced interpretability abilities in the earlier layers of models compared to later ones. Future research could build on our findings by interpreting VJPs using alternative projection methods, such as those proposed by [Din et al. \(2023\)](#), or by employing techniques like sparse autoencoders ([Bricken et al., 2023](#)).

9 Limitations

Our use of LL in projecting gradients has limitations when it comes to explaining the gradients of earlier layers. At this point, it remains unclear whether gradients operate in the same embedding space across all layers or if another transformation is required for projecting earlier layers. This question is currently being explored for the forward pass (see [Section 3.3](#)), suggesting additional learned transformations to the first layers. Given the lack of a wide consensus on this additional transformation, we have opted to employ only the original LL projection in our analysis. Furthermore, some recent contributions against LL argue that this method is more correlated with LMs’ behaviors, rather than causally explaining them. Our work shows that at least in the later layers of LMs, token embeddings are directly placed into the weights of the LM, making LL projections well-justified.